

CoSc 3311

Fundamentals of Software Engineering

Dagmawi Lemma (PhD)

Assistant Professor, AAU

courses.dag@gmail.com

Dagmawi L.
(courses.dag@gmail.com)

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6. Software Project Management

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(courses.dag@gmail.com)

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What is Project?

- A project is a **temporary** endeavor undertaken using limited **resources** to achieve a **specific goal**
 - Temporary: bounded with definite beginning and ending time
 - Ends when the goal is achieved or when it become apparent that the goal will not be achieved or when the achieving the goal is no longer relevant
 - Not necessarily short period – can range from weeks to years
 - The project may last but could have an impact that outlast the project lifetime
 - Resources: cost incurred to acquire material, equipment and labor
 - Specific goal: delivering product or service where the product or service is distinguishable from other product or service

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(courses.dag@gmail.com)

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Examples of IT Project

- Developing Software
- Installing Network Infrastructure
- Constructing Data Center
- Upgrading ICT infrastructure

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Progressive Elaboration

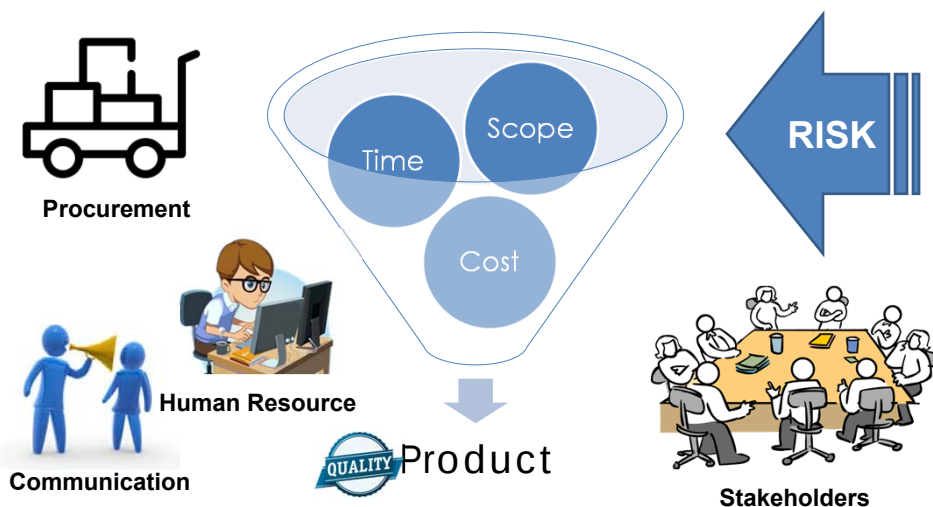
- Progressive elaboration is characteristic of a project.
 - Progressively: proceeding in steps and incrementally
 - Elaborated: worked out in detail and thoroughly
 - For example:
 - System development project begins by identifying the business requirement and setting goals. Then requirements to be satisfied are identified. Once the requirements are approved to be feasible the analysis model is developed and followed by design. Based on the design specification, the system is implemented (constructed).
 - Each phase needs to be well elaborated using models, specifications, diagrams, descriptions, and code.

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What is Project Management?



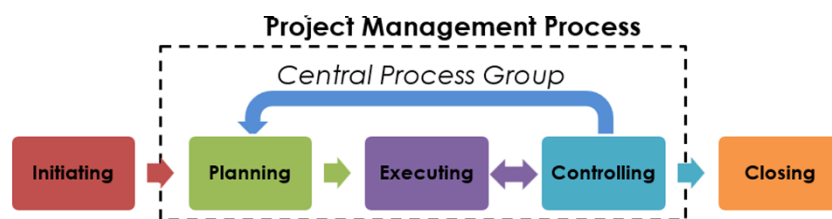
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What is PM? (continued)

- Project Management is the application of knowledge, skills, tools and techniques to project activities with the intention of meeting the project requirements so that the necessary goal can be achieved with the desired quality.
- Project management runs as a group of processes that describe, organize and complete the work of the project.



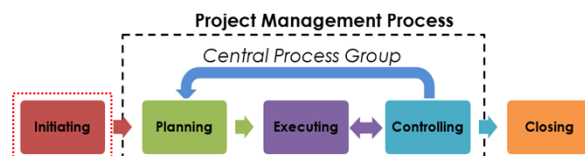
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Project Process Group

- Initiating processes
 - authorizing the project or a **phase**



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Project Process Group

- Planning processes
 - defining and refining objectives
 - selecting and budgeting the required resource with the best of the alternative course of action to attain the project objective



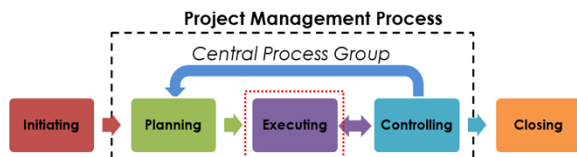
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Project Process Group

- Executing processes
 - coordinating and allocating the required resources to carry out the plan



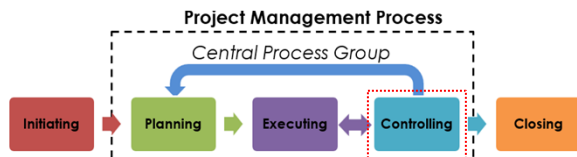
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Project Process Group

- Controlling processes
 - ensuring that the execution is being done as per the plan and the project is running toward its objectives, as well as measuring progress regularly



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Project Process Group

- Closing processes
 - formalizing acceptance of the project or **phase** and bringing it to an orderly end



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Project Phases

- Each project involves or divided into phases so that it is possible to efficiently control and manage the project.
- Each project phase is marked by one or more deliverable.
 - Deliverables are tangible and verifiable artifacts.
 - E.g. feasibility study, detail design, a working prototype, etc.

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(courses.dag@gmail.com)

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Project Phases (Continued)

- Phase-end is the stage where that particular phase mark its deliverable; and the phase end is also called phase exists, kill points, stage gates.
- At phase-end it is important to review:
 - the deliverable to verify if the artifact meets the desired quality
 - the project performance
 - to determine if it is feasible to proceed to the next phase
 - to detect and correct errors cost effectively

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(courses.dag@gmail.com)

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Project Life Cycle

- The project phases collectively forms the project life cycle.
- The project life cycle defines the starting and ending phase of a project
 - For example, should a need assessment or feasibility study be separate project or first phase of the project?
- Product-oriented processes specify and create the project product (e.g. the software process model)
- The project life cycle definition:
 - sets phases of the project and what technical work should be done at each phase
 - describes which transition actions at the beginning and end of the project are included
 - States who is involved at which phase

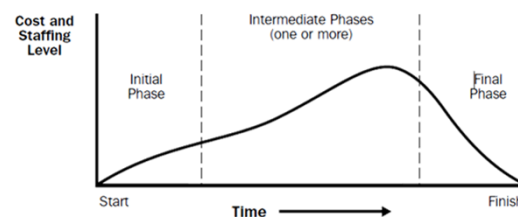
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Project Life Cycle (continued)

- Common characteristics shared by most project life cycle description
 - Cost and staffing is lower at the start but high toward the end



Sample Generic Life Cycle

- The probability of **successfully** completing the project is lowest as the result of risk and uncertainty, which is high at the beginning of the project
- Influence by stakeholders is high at the beginning but gradually decline as the project progress

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(courses.dag@gmail.com)

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Organizational Influence

- Project based organizations
 - Whose operations or job is primarily running projects hence the organizational structure is projectized
 - E.g. software development company, real estate, construction company, design and supervision consultants, etc.
 - organizations adopted *management by project* often has functional structure
 - For example, when a new product development is undertaken in a factory, the design phase of that product is considered as *design project* and is done by the engineering department
- The absence of project oriented system usually makes project management more difficult

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(courses.dag@gmail.com)

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Organizational Influence

Project Characteristics / Organization Structure	Functional	Matrix			Projectized
		Weak Matrix	Balanced Matrix	Strong Matrix	
Project Manager's Authority	Little or None	Limited	Low to Moderate	Moderate to High	High to Almost Total
Percent of Performing Organization's Personnel Assigned Full Time to Project Work	Virtually None	0–25%	15–60%	50–95%	85–100%
Project Manager's Role	Part-time	Part-time	Full-time	Full-time	Full-time
Common Titles for Project Manager's Role	Project Coordinator/ Project Leader	Project Coordinator/ Project Leader	Project Manager/ Project Officer	Project Manager/ Program Manager	Project Manager/ Program Manager
Project Management Administrative Staff	Part-time	Part-time	Part-time	Full-time	Full-time

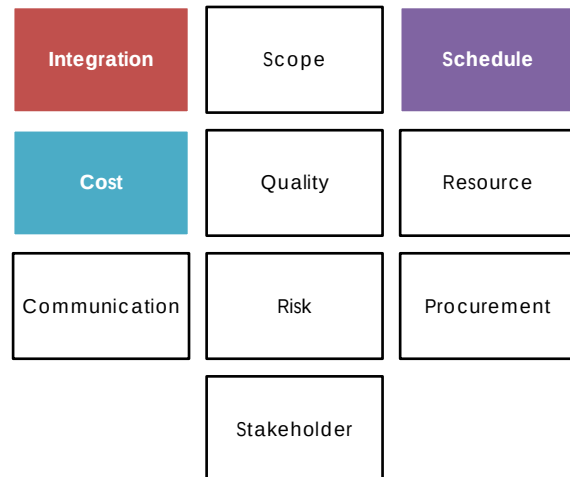
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PM Knowledge Areas

- The knowledge and practices described in PMBOK are generally accepted as they are applicable to most projects.
 - However those knowledge and practices may not be applied uniformly on all projects.
 - For example IT Vs Constructing project : each has its own unique characteristics and constraints, hence the project management team is responsible for determining what is appropriate for their particular project.
- Project management knowledge draws on ten areas.



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(courses.dag@gmail.com)

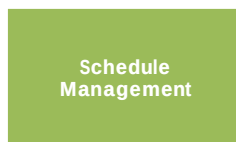
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Selected Topics



Includes identifying, defining, combining, unifying, and coordinating the various processes and project management activities within the Project Management Process Groups.



Includes the processes required to manage the timely completion of the project.



Includes planning, estimating, budgeting, financing, funding, managing, and controlling costs

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(courses.dag@gmail.com)

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Integration Management

- Project Charter – a document that formally authorizes the existence of a project and provides the project manager with the authority to apply organizational resources to project activities.
- Project Management Plan – a document that describes how the project will be executed, monitored and controlled, and closed.

Subsidiary Management Plans

Schedule management plan

Scope management plan

Requirements management plan

Cost management plan

Quality management plan

Resource management plan

Communications management plan

Risk management plan

Procurement management plan

Stakeholder engagement plan

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(courses.dag@gmail.com)

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Project Charter

- The first process during Integration Management is to produce the Project Charter
- Project Charter:
 - Project Charter includes a high level information
 - Project charter can be considered as the identity of the project
 - A Project Charter needs to provide the following information
 - What is it?
 - Why it's useful?
 - How it's used?
- **Note that buyers and sellers should have independent charter**

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(courses.dag@gmail.com)

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Project Charter (continued)

- Project Charter as a simple document comprising, at least
 - Project Name (Title) and Description
 - Formal Authorization-Description (paragraph 1)
 - States the organization, the needs of the organization and the solution this project is about to deliver as well as when it will be done
 - Project Scope (paragraph 2)
 - Describes at a higher level the scope and the major deliverables of the project
 - Project Manager (paragraph 3)
 - Describes the responsibility and the authorization the manager/s has/have in acquiring and assigning the different resources
 - Milestone Schedule (paragraph 4)
 - High-level milestones
 - Project Budget (paragraph 5)
 - The allocated or required total budget
 - And lastly the acceptance section where the sponsors or owner of the project sign on the document

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(courses.dag@gmail.com)

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Project Charter (continued)

- Project Charter, more comprehensively need to document
 - Project Title and Description
 - Project Manager's information and the authority level
 - Business Case (i.e., why the project is needed)
 - Resources (pre assigned though some will be procured in the future)
 - Stakeholders and stakeholders requirements
 - Not that sometimes the requirements might conflict but the agreed and final requirements shall be documented (this might affect the scope)
 - Product Description (may include deliverables)
 - Measurable Project Objectives
 - Project Approval Requirements (when do we say the project is "Done"?)
 - High Level Project Risks (opportunities and threats)
 - Project Sponsor Authorizing the Project (sponsors/owner should sign)

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(courses.dag@gmail.com)

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Project Selection

- Project Managers must know why their project was selected and how it fits into the organization's strategy
- Two approaches
 - Benefit measurement (Comparative Approach)
 - Evaluate and compare several projects based on their benefits, profits, revenue etc.
 - The optimum one is selected
 - Constrained Optimization Method (Mathematical Approach)
 - Takes several inputs about projects and the inputs are mathematically modeled

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(courses.dag@gmail.com)

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Project Selection (Continued)

- Benefit measurement methods
 - Murder board
 - A panel of people who try to shoot down new project would emphasize on the negative aspect of the project and if the project is able to defend it then it will be selected
 - Peer review
 - Project selection is made based on the perspective of several people (experts and stakeholders)
 - Scoring models
 - **Economic Models**

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(courses.dag@gmail.com)

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Project Selection (Continued)

- Economic Model
 - Present Value
 - Net Present Value
 - Internal Rate of Return
 - Payback Period
 - Benefit-Cost ratio

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(courses.dag@gmail.com)

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Project Selection (Continued)

- Economic Model – Present Value
 - Present value calculates the present value of a money that will be paid and over thrived in future
 - The value of a Birr we have today has more value than a Birr in the future
 - Interest rates are used to calculate the PV
 - $PV = \frac{FV}{(1+r)^n}$
 - PV = Present Value
 - FV = Future Value (i.e., cash inflow or outflow)
 - r = interest rate
 - n = number of periods

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(courses.dag@gmail.com)

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Project Selection (Continued)

- Economic Model – Net Present Value
 - NPV is calculated using PV
 - It is the result of the sum of present values over a given period

$$NPV = \sum_{i=0}^n \frac{FV}{(1+r)^i}$$
 - For example if a 300,000ETB worth project has three installments over three years and if in each year 100,000 ETB would be paid and the annual interest rate is 10%
 - 1st installment's PV = $\frac{FV}{(1+r)^n} = \frac{100,000}{(1+0.10)^0} = 100,000$ ETB
 - 2nd installment's PV = $\frac{FV}{(1+r)^n} = \frac{100,000}{(1+0.10)^1} = 90,909$ ETB
 - 3rd installment's PV = $\frac{FV}{(1+r)^n} = \frac{100,000}{(1+0.10)^2} = 82,645$ ETB
 - Total cash flows = 300,000ETB while the NPV = 273,554 ETB
 - A project will be selected if the NPV > 0

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(courses.dag@gmail.com)

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Project Selection (Continued)

- Economic Model – Internal Rate of Return
 - The IRR explains how much percent a project will turn back.
 - If 100, 000 ETB is needed to initiate the project and if the IRR is 12% in 2 years, then the project will bring 12,000 ETB by the end of the second year
 - Project with higher IRR should be selected but in view of the NPV
 - Basically the IRR is the rate at which the NPV becomes zero

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(courses.dag@gmail.com)

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Project Selection (Continued)

- Economic Model – Payback Period
 - The time required to recover the investment of a project
 - Projects with shorter payback period should be selected

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(courses.dag@gmail.com)

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Project Selection (Continued)

- Economic Model – Benefit-Cost Ratio
 - The project cost is compared against the value of the benefit
 - Benefit : Cost > 1 ; i.e., benefit $>$ costs (selected project)
 - Benefit : Cost $= 1$; i.e., benefit = costs
 - Benefit : Cost < 1 ; i.e., benefit $<$ costs

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(courses.dag@gmail.com)

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Project Selection (Continued)

- Other cost considerations
 - Economic Value Added (EVA)
 - Opportunity Cost
 - Sunk Cost

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(courses.dag@gmail.com)

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Project Selection (Continued)

- Other cost considerations - Economic Value Added (EVA)
 - Concerned with whether the project returns to the company more than its costs
 - It is the amount of added value the project produces for the company's shareholders above the cost of project
 - Note that the added values can be financial or intangible

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(courses.dag@gmail.com)

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Project Selection (Continued)

- Other cost considerations – Opportunity Cost
 - Opportunity Cost is the forgone benefit that would have been derived by an option not chosen
 - E.g., suppose the NPV of project A and B is 300,000 ETB and 250, 000 ETB respectively if either of the two has to be selected
 - A has higher NPV so it is worth to be selected, but it forgo the NPV of B hence the opportunity cost of A is 250, 000 ETB

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(courses.dag@gmail.com)

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Project Statement of Work

- Created by customer/sponsor and describes their needs, product scope, and how the project fits into their strategic plan
 - E.g., user requirements specifications (URS) for a software

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(courses.dag@gmail.com)

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Selected Topics

Integration Management

Includes identifying, defining, combining, unifying, and coordinating the various processes and project management activities within the Project Management Process Groups.

Schedule Management

Includes the processes required to manage the timely completion of the project.

Cost Management

Includes planning, estimating, budgeting, financing, funding, managing, and controlling costs

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(courses.dag@gmail.com)

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Schedule Management

- Project Scheduling provides a detailed plan that represents how and when the project phases will be executed and deliver the deliverables that are defined in the project scope
- The Project Schedule serves as a tool for communication, managing stakeholders' expectations, and as a basis for performance reporting.
- Before we start scheduling, we need to know and decide the scheduling approach
 - Iterative scheduling
 - Requirements are documented as user stories (product backlog), which is then prioritized. Then the features of each product are developed with specific timeframe (1-4weeks) (known as sprint)
 - On-demand scheduling
 - Does not rely on predefined schedule, but pulls product from the backlog as needed

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(courses.dag@gmail.com)

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Things we do during schedule management

- Plan Schedule Management
- Define Activities
- Sequence Activities
- Estimate Activity Durations
- Develop Schedule
- Control Schedule

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Planning Schedule Management

- Planning schedule management is the process of establishing the policies, procedures, and documentation for planning developing, managing, executing, and controlling the project schedule
 - Inputs are the project charter, project management plan, enterprise environmental factors, and organizational process assets
 - Uses expert judgment, data analysis, and meetings as method

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(courses.dag@gmail.com)

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Schedule Management Plan

- Planning schedule management results in a plan that establish
 - Project schedule model development
 - Release and iteration length (e.g., sprint length)
 - Unit of measures (e.g. person-hours, or person-month)
 - Person-**hours**, for example, is effort needed to do a task by a person without interruption

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(courses.dag@gmail.com)

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Defining and Sequencing Activities

- The WBS is created during scope management, and now it can be used in the process of identifying and documenting the specific actions to be performed to produce the projects deliverable
- The leaf nodes of the WBS can be used to estimate the time, resource and cost
 - Or work packages can be further decomposed from the leaf nodes

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(courses.dag@gmail.com)

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Activity Attributes

- Defining activities also involves identifying and describing activity attributes, which are additional information about each activity and useful in scheduling
 - the dependency between activity can be used in sequencing the activities and setting the predecessor task in the Gantt Chart
- Activity attributes includes but are not limited to:
 - Unique identifier (ID) of the activity
 - The WBS ID
 - Activity label or name
 - Dependency (predecessor activity/ies)
 - Estimated time
 - Etc.

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(courses.dag@gmail.com)

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Milestones

- A milestone is a significant point or event in a project with **zero** duration.
 - For example a date set for delivering an intermediate product
- Mandatory milestones are those which are required in the contract
 - For example, delivering the *Requirement Specification Document* is possibly mandatory but *Documenting the Interview Transcript* is not mandatory milestone though it could be one of the important activity in the Requirement Analysis phase of a system development project

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(courses.dag@gmail.com)

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Sequencing Activities

- Having the list of activities and their attributes, the activities should be sequenced in their logical order by identifying and documenting the relationship between the activities
- Precedence diagram method (PDM) represents the sequence of activities using directed graph, where the nodes represent the activity and the edges represent the logical order of activities

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Sequencing Activities (Continued)

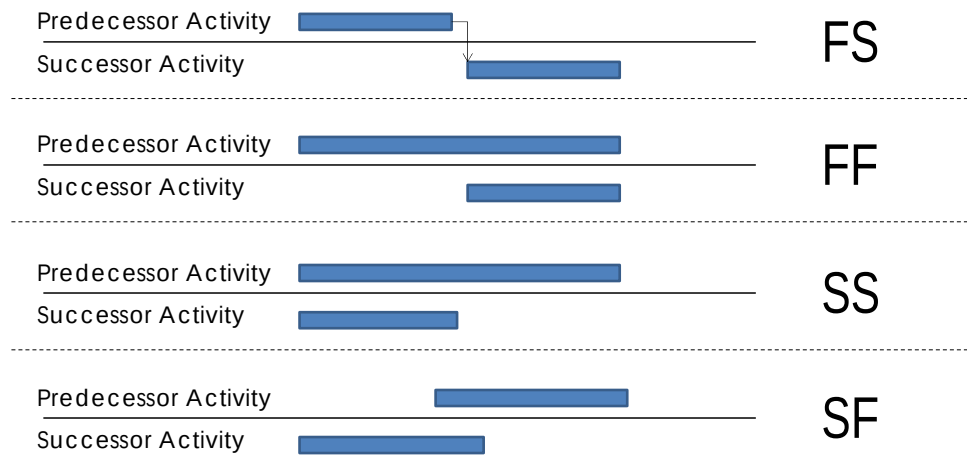
- Four types of dependencies can be considered in PDM
 - Finish-to-Start (FS)
 - A logical relationship in which a successor activity cannot start until a predecessor activity has finished.
 - For example, installing an application in the deployment environment (successor) cannot start until the deployment environment is ready (e.g. OS) (predecessor).
 - Finish-to-Finish (FF)
 - A logical relationship in which a successor activity cannot finish until a predecessor activity has finished.
 - For example, documenting requirement (successor) cannot be completed until the requirement analysis is completed
 - Start-to-Start (SS)
 - A logical relationship in which a successor activity cannot start until a predecessor activity has started.
 - For example, package diagrams (successor) cannot be started until the system decomposition is started
 - Start-to-Finish (SF)
 - A logical relationship in which a successor activity cannot finish until a predecessor activity has started.

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(courses.dag@gmail.com)

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Sequencing Activities (Continued)



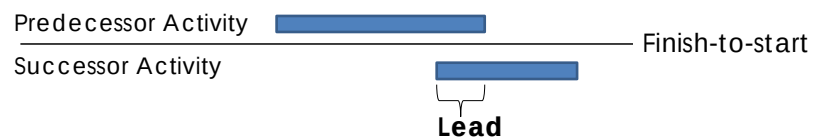
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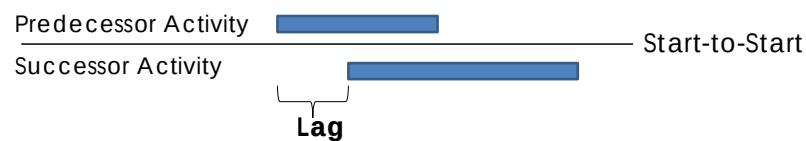
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Leads and Lags

- Lead is the amount of time a successor activity can be advanced with respect to a predecessor activity



- Lag is the amount of time a successor activity will be delayed with respect to predecessor activity



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(courses.dag@gmail.com)

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Estimating Duration

- After the sequence and the dependency between activities is defined, it is important to estimate the required duration for each activity
 - Note that the estimation is done throughout the project
- Information required to estimate the duration of an activity includes, but are not limited to:
 - scope of the work,
 - required resource types or skill level,
 - estimated resource quantity (requires expert judgment in some projects like software development)
- Also it is important to determine the **critical path** of the project
 - This, however, is related with activity sequencing as well

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(courses.dag@gmail.com)

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Factors in Estimating Duration

- Law of Dimensioning Returns
 - If we plan to crashing the project schedule, we might need to add more resources. However, the addition of resource might not linearly improve the productivity and the return we get might deteriorate.
- Number of Resources
- Advances in Technology
- Motivation of Staff

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(courses.dag@gmail.com)

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Techniques in Estimating Duration

- Parametric
 - Estimation is calculated based on historical data and project parameter
- Three-Point
 - Estimation is done by considering uncertainties and risk and taking an approximate range for an activity's duration
 - For each activity the most-likely (tM), optimistic time (tO) and pessimistic time (tP) are estimated and the average is taken as estimated time (tE)

$$tE = \frac{tM + tO + tP}{3}$$
- Bottom-up
 - Estimation is made by aggregating the estimate of the lower-level tasks in the WBS

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(courses.dag@gmail.com)

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PERT Chart

- Planning using PERT/CPM Chart
 - Define activities in a project and identify their dependency (E.g. consider the following list of activity:)

Activity No.	Activity	Expected Completion Time	Dependency
1	Requirement Identification	5	-
2.	Requirement Structuring	6	1
3.	UI Design	7	1
4.	Database Design	2	2,3
5.	Documentation	6	4
6.	Programming	5	4
7	Testing	3	6
8	Installation	1	5,7

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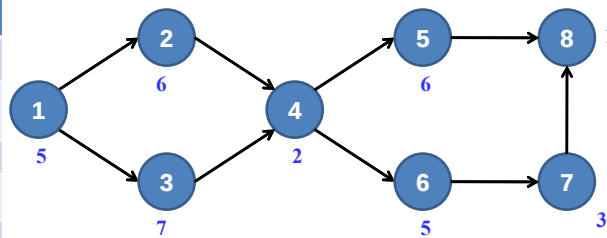
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Project Management and Planning

• Planning using PERT/CPM Chart

- In the PERT chart the nodes indicate the activities while the arrows indicates the time path

No.	Activity	Exp. Time	Dep.
1	Req. Identification	5	-
2	Requirement Structuring	6	1
3	UI Design	7	1
4	Database Design	2	2,3
5	Documentation	6	4
6	Programming	5	4
7	Testing	3	6
8	Installation	1	5,7



Using information from the table,
indicate expected completion time for
each activity

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(courses.dag@gmail.com)

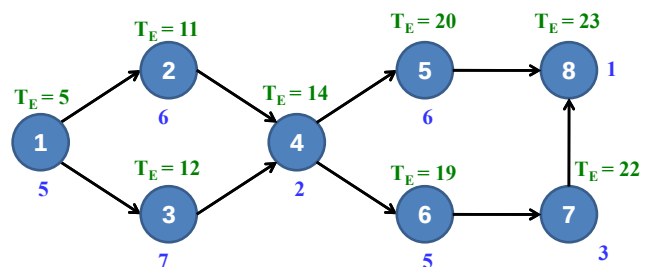
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Project Management and Planning

• Planning using PERT/CPM Chart

- Hint: the earliest expected completion time for a given activity is determined by summing the expected completion time of this activity and the earliest expected completion time of the immediate predecessor.
- Rule: if two or more activities precede an activity, the one with the largest T_E is used in calculation (e.g., for activity 4, we will use T_E of activity 3 but not 2 since $12 > 11$)
- After indicating expected completion time for each activity, Calculate earliest expected completion time (T_E) for each activity and finally the T_E for the entire project.



Dagmawi L.
(courses.dag@gmail.com)

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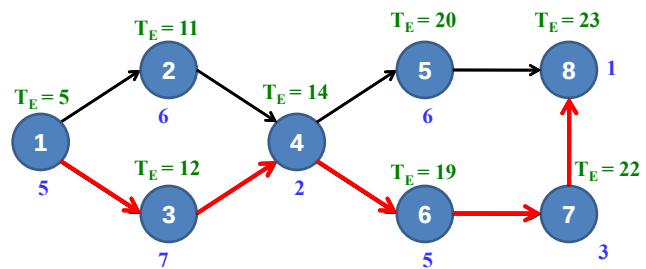
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Project Management and Planning

• Planning using PERT/CPM Chart

– **Critical Path**: The critical path represents the *shortest* (Practical) time, in which a project can be completed.

- Any activity on the critical path that is delayed in completion, delays the entire project
- Activities not on the critical path contain **slack time** and allow the project manager some flexibility in scheduling

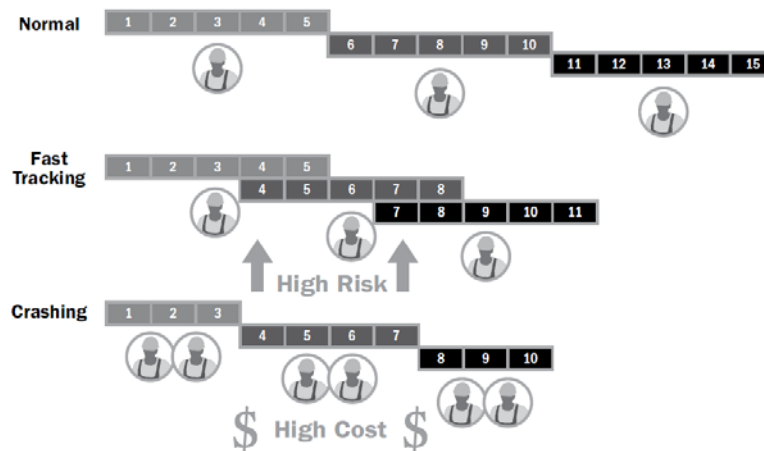


Dagmawi L.
(courses.dag@gmail.com)

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Schedule Compression



Dagmawi L.
(courses.dag@gmail.com)

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Selected Topics

Integration Management

Includes identifying, defining, combining, unifying, and coordinating the various processes and project management activities within the Project Management Process Groups.

Schedule Management

Includes the processes required to manage the timely completion of the project.

Cost Management

Includes planning, estimating, budgeting, financing, funding, managing, and controlling costs

Dagmawi L.
(courses.dag@gmail.com)

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Cost Management

- Cost management includes planning, estimating, budgeting, financing, funding, and controlling costs with the objective of enabling the project to be completed within approved budget

Dagmawi L.
(courses.dag@gmail.com)

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What is Cost?

- Whatever expended in monetary term to satisfy the needs (e.g. human resource, martial resource, space, administration needs, etc.) in the project is cost
- Cost can be, *at least*, of three types
 - Direct cost
 - Indirect Cost
 - Contingency Cost

Dagmawi L.
(courses.dag@gmail.com)

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Direct Cost

- Direct Cost in a project refers to what is expended or to be expended in connection to the product or the process that is stated in the Scope Statement
 - Example
 - Software license, professional fee, travel (e.g. for data collection), testing (test user allowance, test lab construction /lease), etc.
- NOTE:
 - the cost item that is considered as direct cost may not be always a direct cost
 - it rather shall be categorized in connection with the Scope Statement

Dagmawi L.
(courses.dag@gmail.com)

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Indirect Cost

- Indirect Cost in a project refers to what is expended or to be expended while using shared resources with other projects or activities that are not directly related but support the process which is stated in the Scope Statement
 - Example
 - Office space, general administration cost, office equipment, refreshments, etc.
- NOTE:
 - the cost item that is considered as indirect cost may not be always an indirect cost.
 - it rather shall be categorized in connection with the Scope Statement

Dagmawi L.
(courses.dag@gmail.com)

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Indirect Cost (Continued)

- There are two types of indirect costs.
 - fixed indirect costs which contains activities or costs that are fixed for a particular project or company
 - like transportation of labor to the working site,
 - building temporary building (like project office), etc. T
 - recurring indirect costs which contains activities that repeat for a particular company
 - Accounting service (e.g., payment of salaries)
 - Utility cost, etc.

Dagmawi L.
(courses.dag@gmail.com)

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Contingency Cost

- When estimating, there is always uncertainty as for the precise items or contents, hence the cost estimation might face uncertainty; and that is a risk to the project
 - Such risk are also known-unknowns
 - Known from past experience
 - Unknown whether to occur or not
 - The known-unknown cost is also Cost Contingency
- Cost Contingency is estimated often based on past experience, but
 - Should not be taken for granted
 - Should not take significant portion of the total project cost
 - Often better not to exceed more than 10%

Dagmawi L.
(courses.dag@gmail.com)

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Contingency (*continued*)

- Cost Contingency shall exclude
 - Major changes in the product specification
 - Extraordinary events (force majeure)
 - A typical force majeure clause permits a party to extend, suspend and/or terminate the performance of the contract when an extraordinary event occurs beyond the parties' control
 - Management reserves
 - is an amount of contract budget set aside for **management** control purposes (known unknowns) rather than designated for the accomplishment of one or more tasks
 - Escalation and currency effects
 - inflation

Dagmawi L.
(courses.dag@gmail.com)

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Contingency (*continued*)

- Contingency is estimated using statistical analysis and/or based on past experience
 - Shall be expended
 - Methods used to estimate
 - Expert judgment
 - Predetermined guidelines
 - Simulation analysis
 - Monte Carlo
 - Parametric modeling – usually use regression analysis
- In risk management, contingency is considered as cost control

Dagmawi L.
(courses.dag@gmail.com)

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Processes in Cost Management

- The Project Cost Management processes involves
 - Planning cost management
 - Estimating costs
 - Determining and allocating budget
 - Control costs

Dagmawi L.
(courses.dag@gmail.com)

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Planning Cost Management

- Before estimating, defining and controlling costs, it is important to decide/agree on how the project will be estimated, budgeted, managed and controlled.
- Therefore, the key importance of Planning Cost Management is to provide guidance and direction on how the project cost is managed throughout the project

Dagmawi L.
(courses.dag@gmail.com)

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Estimating Cost

- Estimating the cost will help us to determine the monetary resources needed to complete the project.
- Cost estimation is a rationally made prediction based on information at a given point in time.
- The information (inputs) needed to estimate the cost for each activity includes:
 - Project management plan
 - Cost and quality management plan, Scope baseline
 - Project documents
 - Lessor learned register, project schedule, resource requirements, risk register
 - Enterprise Environmental factor
 - Market condition, published commercial information, exchange rates and inflation
 - Organizational process assets
 - Cost estimating policies, cost estimating templates, historical information's (lesson learned repository)

Dagmawi L.
(courses.dag@gmail.com)

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Estimating Cost (continued)

- Depending on the type of the project, the tools and techniques used to estimate cost might vary.
 - For example, if most of the metrics are known, then parametric approach can be followed, but this may not be applicable for all cases.
- Similar to resource and schedule estimation expert judgment, analogues estimating as well as parametric estimating can be considered as applicable.

Dagmawi L.
(courses.dag@gmail.com)

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Estimating Cost (continued)

- Bottom-up estimating
 - The cost of individual works are estimated in detail and the detailed costs are summarized
- Three-point estimating
 - This method considers the most likely cost (cM), optimistic cost (cO), and pessimistic cost (cP) and computes the estimated cost (cE) using triangular distribution or beta distribution
 - Triangular distribution $cE = \frac{cM+cO+cP}{3}$
 - Beta distribution $cE = \frac{4cM+cO+cP}{6}$

Dagmawi L.
(courses.dag@gmail.com)

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Estimating Cost (continued)

- Once the cost estimation is completed, the summary shall be prepared including all necessary costs such indirect cost and cost contingencies

Dagmawi L.
(courses.dag@gmail.com)

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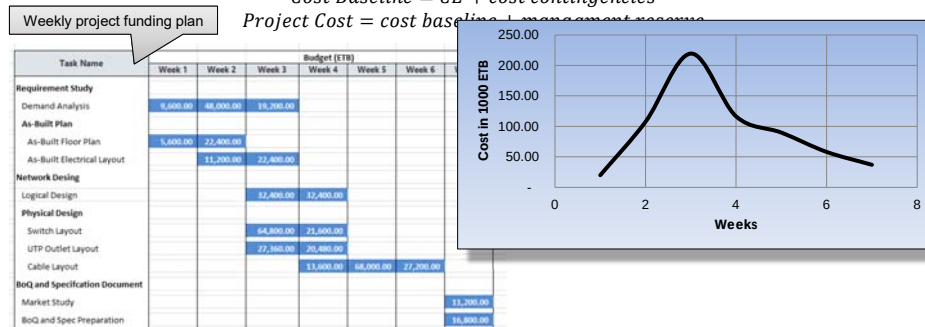
Determining Budget

- The estimated costs (direct, indirect and contingencies) of individual activities need to be aggregated as budget
- Determining the budget is important to determine the cost baseline against which project performance can be monitored and controlled.
- The cost baseline is the time-phased project budget that should be aligned with the project schedule.

$$\text{Cost Estimate of Work Package (CE)} = \text{Direct Cost} + \text{Indirect Cost}$$

$$\text{Cost Baseline} = \text{CE} + \text{cost contingencies}$$

$$\text{Project Cost} = \text{cost baseline} + \text{management reserve}$$



Dagmawi L.
(courses.dag@gmail.com)

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Controlling Cost

- Once the project is started and costs are being expended, it is important to monitor the status of the project and follow-up the cost performance.
 - This will be done by analyzing the cost baseline against the actual cost being spent
 - Earned Value Analysis (EVA), can be used to compare the performance measurement baseline to the actual schedule and cost performance
 - It uses earned value (EV), which is also known as budgeted cost of work performed (BCWP)
 - And the actual cost (AC), which is the actual cost of work performed (ACWP)

Cost performance index (CPI) is a ratio of EV:AC

- If $CPI < 1$ the project is over budget
- If $CPI > 1$ the project is under budget

$$CPI = \frac{BCWP}{ACWP}$$

Cost variance (CV) is the difference between the EV and AC

- If $CV < 0$ the project is over budget
- If $CV > 0$ the project is under budget

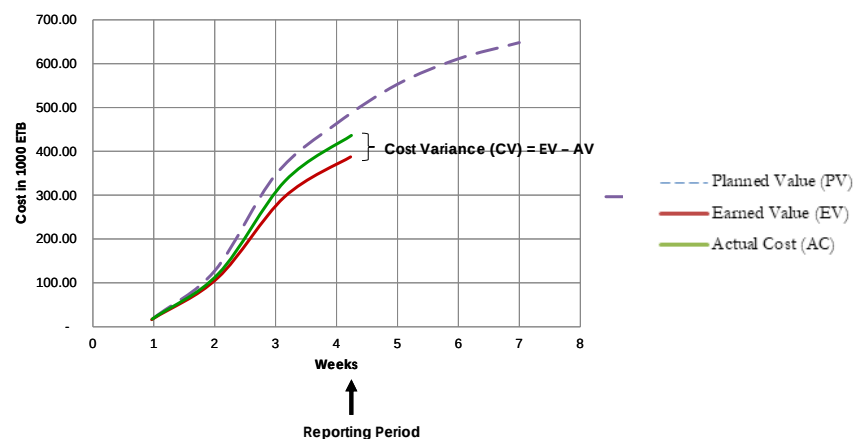
$$CV = BCWP - ACWP$$

Dagmawi L.
(courses.dag@gmail.com)

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Controlling Cost (continued)



Dagmawi L.
(courses.dag@gmail.com)

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